Perplexity AI Analysis of Con Edison EIOPs 25-E-0072 and 25-G-0073

Analysis of Line 589 on DMM Website Exhibits EIOP 1-14

Question # 1

List all examples where Con Ed documents issues related to equipment availability

EIOPs.pdf

Con Edison documents equipment availability or material availability issues in several repeated ways throughout the EIOPs, typically under "Project Risks and Mitigation Plan" and occasionally in work descriptions and justifications.

Below is a list of all distinct examples where Con Edison explicitly flags equipment/material availability, supply chain, lead times, or similar constraints that affect availability of equipment needed for projects or operations.

- "Supply chain issues" as a project risk, with mitigation involving early engagement with Supply Chain and managing needed versus available material across programs, in the Avenue A / Cooper Square / Seaport 2 transfer project.

"Material Availability Issues" as a project risk for the Brooklyn Clean Energy Hub Scalable Reliability project, with mitigation via coordination with Supply Chain and vendors to align with lead times and adjust plans as shortages are encountered.

"Material Availability Issues" as a project risk for the Brooklyn Energy Hub Positions 2S and 2N project, with the same mitigation approach of Supply Chain coordination and vendor engagement around lead times.

"Material Availability Issues" cited as a project risk in at least one Underground Transmission program (e.g., Underground Transmission Structure Modernization / cathodic protection / rectifier programs), where Con Edison notes vendor lead times and the need to stay engaged with vendors and adjust if shortages occur.

Repeated references to "supply chain disruptions," "manufacturing or supply chain disruptions," or similar, in the Transformer Purchases and related Distribution Equipment programs, noting:

- Increasing lead times and decreased available supply for transformers and related equipment.
- Need to increase orders and on-hand inventory of distribution equipment.
- Risk that manufacturing or supply chain disruptions impact procurement costs and equipment availability.

Explicit discussion that custom 4 kV unit substation switchgear has “increasingly long wait times” for custom design and manufacture, and that specialized technicians to install/service this legacy gear are “in increasingly scarce supply,” all of which drive a change to modular, off-the-shelf switchgear to improve equipment availability.

Discussion under the Transformer Purchases / Transformer Installation programs that:

- Customer schedules require transformers to be available before they can be acquired on the market.
- Growing demand for certain equipment has strained global supply.
- Con Edison must secure equipment further in advance of the known need to ensure availability.

References in multiple project-risk sections that growing numbers of projects “may strain existing resources” (engineering, labor, construction) and that Con Edison must expand hiring and pre-plan outages and resources; while not strictly material availability, these are framed as availability of necessary resources to execute projects on schedule.

If you want, a next step can be a pinpointed page/Exhibit list (e.g., “Exhibit EIOP-7 p. xx”) for each bullet above so you can cite them directly.

Question #2

List all examples where Con Ed documents issues related to equipment availability

EIOPs.pdf

Con Edison documents equipment availability or material availability issues in several repeated ways throughout the EIOPs, typically under “Project Risks and Mitigation Plan” and occasionally in work descriptions and justifications.

Below is a list of all distinct examples where Con Edison explicitly flags equipment/material availability, supply chain, lead times, or similar constraints that affect availability of equipment needed for projects or operations.

- “Supply chain issues” as a project risk, with mitigation involving early engagement with Supply Chain and managing needed versus available material across programs, in the Avenue A / Cooper Square / Seaport 2 transfer project.

“Material Availability Issues” as a project risk for the Brooklyn Clean Energy Hub Scalable Reliability project, with mitigation via coordination with Supply Chain and vendors to align with lead times and adjust plans as shortages are encountered.

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“Material Availability Issues” cited as a project risk in at least one Underground Transmission program (e.g., Underground Transmission Structure Modernization / cathodic protection / rectifier programs), where Con Edison notes vendor lead times and the need to stay engaged with vendors and adjust if shortages occur.

Repeated references to “supply chain disruptions,” “manufacturing or supply chain disruptions,” or similar, in the Transformer Purchases and related Distribution Equipment programs, noting:

- Increasing lead times and decreased available supply for transformers and related equipment.
- Need to increase orders and on-hand inventory of distribution equipment.
- Risk that manufacturing or supply chain disruptions impact procurement costs and equipment availability.

Explicit discussion that custom 4 kV unit substation switchgear has “increasingly long wait times” for custom design and manufacture, and that specialized technicians to install/service this legacy gear are “in increasingly scarce supply,” all of which drive a change to modular, off-the-shelf switchgear to improve equipment availability.

Discussion under the Transformer Purchases / Transformer Installation programs that:

- Customer schedules require transformers to be available before they can be acquired on the market.
- Growing demand for certain equipment has strained global supply.
- Con Edison must secure equipment further in advance of the known need to ensure availability.

References in multiple project-risk sections that growing numbers of projects “may strain existing resources” (engineering, labor, construction) and that Con Edison must expand hiring and pre-plan outages and resources; while not strictly material availability, these are framed as availability of necessary resources to execute projects on schedule.

If you want, a next step can be a pinpointed page/Exhibit list (e.g., “Exhibit EIOP-7 p. xx”) for each bullet above so you can cite them directly.

Please Pinpoint page/Exhibit list documenting bullet points above

Here is a pinpoint list by bullet, with Exhibit and page references (using the page tags embedded in the PDF). All cites are to your attached EIOPs file.

- “Supply chain issues” risk and mitigation (Avenue A / Cooper Square / Seaport 2 load transfers):

  Exhibit EIOP-7, p. 16 (“Project Risks and Mitigation Plan – Risk 1 Supply chain issues …”).

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“Material Availability Issues” risk and mitigation (Brooklyn Clean Energy Hub – scalable reliability project):

- Exhibit EIOP-7, p. 21 (“Risk 2 Material Availability Issues … Engineering will work with Supply Chain…”).

“Material Availability Issues” risk and mitigation (Brooklyn Energy Hub Positions 2S and 2N):

- Same risk template and wording as above, in the Brooklyn Energy Hub Positions 2S/2N project section that immediately follows BCEH scalable reliability (see Exhibit EIOP-7, pp. 22–24 for project context and risk section referencing supply chain and vendor lead times).

Underground Transmission / cathodic protection rectifier program – material availability and extended lead times:

- Exhibit EIOP-8, p. 155–157 block for “Non-Network Reliability” shows general reliability program context, while the explicit “Material Availability Issues. Recent Supply Chain issues are causing increased lead times for rectifiers and anodes.” appears in the cathodic protection/rectifier discussion under “Project Risks and Mitigation Plan – Risk 1 Material Availability Issues.”

Transformer and distribution equipment supply-chain / availability language (lead times, constrained global supply, need to increase orders and inventory):

- These statements appear in the Transformer Purchases and related Distribution Equipment programs within the Electric Operations Distribution Engineering exhibits (same EIOPs PDF), where Con Edison discusses increased global demand, longer lead times, and the need to order equipment earlier and hold more inventory; these programs are grouped with other distribution equipment capital programs in Exhibit EIOP-8 (mid-exhibit pages around the general Distribution Equipment / Transformer Purchase narratives).

Custom 4 kV unit substation switchgear – “increasingly long wait times” and “scarce” specialized technicians; rationale for modular gear to improve availability:

- Described in the distribution engineering section for legacy 4 kV unit substation upgrades, within Exhibit EIOP-8, in the text explaining why legacy custom switchgear is being retired in

favor of modular, off-the-shelf equipment; this appears in the narrative portion preceding the detailed funding tables for that 4 kV program (mid-exhibit pages in EIOP-8).

Transformer purchase/installation programs – need to secure transformers earlier given constrained supply and growing demand:

- Same Transformer Purchases / Distribution Equipment program narrative cited above in Exhibit EIOP-8, which notes that customer schedules require transformer availability ahead of acquisition, that global demand is straining supply, and that Con Edison must place orders well in advance to assure availability.

General "delays due to resource support coordination" / "may strain existing resources" (engineering, labor, construction) in large capital programs:

- Brooklyn Clean Energy Hub Scalable Reliability: Exhibit EIOP-7, p. 21 ("Delays due to resource/support coordination… large number of projects… may strain existing resources.
- Similar "Delays due resources support coordination" language also appears in multiple other project risk sections, including:
- Elmsford – Add 138 kV Disconnect Switches: Exhibit EIOP-8, pp. 79–81.

Oil Minder / Sump Pump program (Substation vault water/oil management): Exhibit EIOP-12, pp. 7–8 ("Delays due resource support coordination").

Williamsburg Network Improvements – explicit "Material availability with supply chain issues" and mitigation:

- Exhibit EIOP-7, p. 304 ("Project Risks and Mitigation Plan – Risk 1 Material availability with supply chain issues…" and mitigation via working with manufacturers, Stores, and Supply Chain to maintain inventory and anticipate requirements).

Oil Minder / sump-pump vault program – "Material Availability Issues" risk and mitigation:

- Exhibit EIOP-12, p. 8 ("Risk 2 Material Availability Issues … Engineering will work with Supply Chain to establish a cohesive plan that aligns with vendor lead times.").

Underground transmission cathodic protection / rectifier program – explicit "Material Availability Issues" and supply chain lead times:

- Exhibit EIOP-12, near p. 5 (within the "Project Risks and Mitigation Plan – Risk 1 Material Availability Issues. Recent Supply Chain issues are causing increased lead times for rectifiers and anodes.").



Engineers Transform Transformers to Save the Power Grid

A global shortage of this esΣΣsential grid tech bottlenecks energy projects

By Andrew Moseman

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SILICON RANCH

At a solar facility in Georgia, Silicon Ranch stores substation transformers it expects to deploy in the region in one to two years.

To Nick de Vries, chief technology officer at the solar-energy developer Silicon Ranch, a transformer is like an interstate on-ramp: It boosts the voltage of the electricity that his solar plants generate to match the voltage of grid transmission lines. "They're your ticket to ride," says de Vries. "If you don't have your high-voltage transformer, you don't have a project."

Recently, this ticket has grown much harder to come by. The demand for transformers has spiked worldwide, and so the wait time to get a new transformer has doubled from 50 weeks in 2021 to nearly two years now, according to a report from Wood MacKenzie, an energy-analytics firm. The wait for the more specialized large power transformers (LPTs), which step up voltage from power stations to transmission lines, is up to four years. Costs have also climbed by 60 to 80 percent since 2020.

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The Great Electrician Shortage

Going green will depend on blue-collar workers. Can we train enough of them before time runs out?

By David Owen

April 24, 2023

Two days before Christmas, rain and high winds knocked down power lines on our road, in a small town in northwestern Connecticut, and that night the temperature dropped to the single digits. I worried that the pipes in our baseboard hot-water radiators would freeze and burst, so at four in the morning I left my wife and our dog shivering in bed, groped my way down to the basement, and, with help from YouTube, attempted to drain the system. Here's a home-improvement tip: if you think that someday you might need to perform an emergency maintenance chore, study it on a summer afternoon when you're not wearing pajamas and a headlamp while trying to hold your cell phone and a bucket.

I did succeed in removing many gallons of water, but when the power came back, thirty hours after it had gone out, I couldn't get the heat going again. I left messages for several plumbers. Pipes had frozen all over the Northeast, so I worried that none would call me back, but then one did: Marc LeMieux, who came over the day after Christmas and showed me what I'd been doing wrong. I was lucky to get him; he told me that in recent years he'd been so overwhelmed by other plumbing work that he'd stopped servicing heating systems. "There aren't enough plumbers now, Dave," he said. "What do you think it's going to be like in ten years?"

Many skilled trades face similar shortages, and those shortages have environmental consequences. The Inflation Reduction Act includes billions in tax credits and direct funding for a long list of climate-friendly projects, but all of them depend on the availability of workers who can execute and maintain them. Last year, on Ezra Klein's *Times* podcast, my colleague Bill McKibben said, "If you know a young person who wants to do something that's going to help the world and wants to make a good living at the same time, tell them to go become an electrician." This seems logical—you can't electrify without electricians—but it doesn't fully describe the need. My daughter and her husband hired an electrician to install an outlet next to their driveway, for their plug-in hybrid minivan, but the car, its network of charging stations, and the electric grid itself wouldn't exist without welders, machinists, mechanics, carpenters, pipe fitters, and many others. In new construction, electric heat pumps are rapidly becoming the default option, for both heating and cooling, but on most installations the bulk of the work is done not by electricians but by heating, ventilation, and air-conditioning (*HVAC*) techs. Plumbers are indispensable, too. Changing weather patterns and rising sea levels threaten access to clean water in many parts of the country, and when water infrastructure fails entire communities suffer, as in the ongoing crises in Flint, Michigan, and Jackson, Mississippi. Plumbers also work on many energy-related projects, including the installation of ground-source heating-and-cooling systems. According to a recent report published by Associated Builders and Contractors, a trade group, job openings in the construction industry averaged three hundred and ninety thousand a month in 2022, and the shortfall was made more ominous by the fact that roughly a quarter of existing workers are older than fifty-five.

One reason for the skilled-labor gap is that the work is real work. The electricians who restored power to the houses on our road spent Christmas Eve in bucket trucks, buffeted by winds so strong they made the screens on our porch hum like kazoos. LeMieux told me that he's had apprentices who quit after a few months because they had decided the job was too wet, too messy, too cold, too dirty, too hot. A more significant factor may be that, for decades, employers, educators, politicians, and parents have argued that the only sure ticket to the good life in America is a college degree. People who graduate from college do earn more, on average, than people who don't, but the statistics can be misleading. Many young people who start don't finish, yet still take on tens of thousands in education loans—and those who do graduate often discover that the economic advantage of holding a degree can be negated, for years, by the cost of having acquired it.

Those who skip college frequently do better, and not just at first. "One of my guys came to me from the same trade school I went to," LeMieux told me. "He had a couple of friends who went to college, and when they got out they were two hundred thousand dollars in debt and didn't have jobs, and he was already making enough to buy a nice new vehicle and a house. I pay him a good hourly wage, he has health insurance and a 401(k), and he gets holidays, vacation time, and personal days. And he will always work—always." According to the Bureau of Labor Statistics, the annual mean wage for plumbers and electricians is about sixty-three thousand dollars, or roughly the same as that for high-school teachers (who typically need not just college but also a master's degree) and journalists.

At my house, LeMieux was able to restore two baseboard zones but not the one on the ground floor, which had indeed frozen. He told me that, even though I hadn't drained it properly, I had possibly removed enough water so that the ice, when it did form, had had room to expand inside the pipes, rather than causing the copper to rupture—though we couldn't be sure until things warmed up. A few days later, when the temperature had risen back to the mid-forties, I tried what I'd watched him do: I attached a hose to the purge valve on the ground-floor return line, next to the boiler, then goosed the manual water feed. Nothing happened at first, but then, suddenly, water and bits of ice were spewing from the far end of the hose. I e-mailed LeMieux to say the heat was on again, and he wrote back to tell me I was hired.

American public high schools began offering vocational training in a serious way a little more than a century ago. The main goal, usually, wasn't to broaden the abilities of all students but to sequester certain unwelcome newcomers: kids who had grown up on farms, kids whose parents were immigrants, kids who weren't white. Jeannie Oakes, a professor emerita at U.C.L.A., in her book "Keeping Track," which was first published in 1985, describes trade-oriented high-school courses as "usually taught to fairly homogeneous groups of students seen as low achieving or low ability." This is often still true, although the issue is moot in many school districts, in which budget cuts and a focus on college preparation have reduced or eliminated traditional vocational offerings.



The decline in trade education poses a threat to the country's emerging climate policy. Leah Stokes, who helped create the Inflation Reduction Act, said, "We have to change the culture around the importance of these jobs."

A significant trend in recent years, at all levels of education, has been a growing emphasis on so-called *STEM* instruction. The acronym stands for science, technology, engineering, and mathematics—disciplines that, in the words of the U.S. Department of Education, impart "21st century career readiness and global competitiveness" (unlike the useless old humanities). I live about a hundred miles north of New York City. The *STEM* curriculum at the regional public high school that serves my town includes career-oriented classes in agricultural sciences—this area is largely rural—but only a smattering in traditional trades. Leah Stokes, a professor of environmental politics at the University of California, Santa Barbara, who was heavily involved in the creation and passage of the I.R.A., told me, "I don't feel we got enough about workforce development in the bill. We have to change the culture around the importance of these jobs, which are going to be linchpins in the clean-energy transition."

My state has a network of dedicated public vocational schools, called the Connecticut Technical Education and Career System (*CTECS*), which might serve as a national model. It consists of seventeen diploma-granting high schools, two aviation-maintenance schools for adults, an after-school program for juniors and seniors enrolled at conventional high schools, and night classes for people of all ages who already work in trades. "Between eighty-five and ninety per cent of all apprentices in the state come from our district," Pat Ciarleglio, who holds three trade licenses and is the head of apprenticeship education at *CTECS*, told me. "We even get electrical engineers who have done all their formalized university education but decide, Hey, I don't want to sit behind a desk." No other state has anything quite like the Connecticut system. It's overseen not by local school boards but by a single, independent state agency, whose director is appointed by the governor. Funding for the schools comes directly from the state—there are no local budget meetings at which angry parents complain about Judy Blume books in the libraries.

I visited three of the schools in early March, beginning with Eli Whitney Technical High School, in Hamden. I crossed the campus with Brent McCartney—who worked as a union carpenter before he joined the system, first as an instructor and now as a consultant—to see a project financed by Connecticut's electric utilities: the construction of a small house on an elevated site next to the school's athletic fields. All the work was being done by students. The windows weren't in yet, but most of the roof had been framed and the walls sheathed with panels that had integrated moisture and air barriers. "When they insulate, they're going to do a really good job on some parts and a really bad job on others, using a variety of materials," McCartney said. "Then they'll use thermal-imaging equipment to do an energy audit, and they'll come up with solutions for the problems they find." Because the house is a teaching project, one class often disassembles something that another class recently assembled, then assembles it again.



The woodshop at Howell Cheney Technical High School.



A “high-efficiency home” being built at another *CTECS* school.

We were joined by seven plumbing students, who were returning from lunch. Victor Leduc, a junior, told me, “This is our e-house—also known as a high-efficiency home. We plan on

installing a lot of appliances and fixtures that save energy and water." He was wearing construction boots, kneepads, an orange hard hat, and a tool belt. "Buildings are an important cause of CO_2 emissions," he continued. "I don't think that's something that many people look at, but homes like this cut that a lot." The finished house will have solar thermal, solar electric, and geothermal systems, all installed by the students.

High schools in the *CTECS* system alternate academic and vocational classes, on a rotating schedule, roughly two weeks of each at a time. During the vocational segments, the student crews are able to work away from the campus, on jobs contracted with the school. That week, Leduc's plumbing crew, in addition to working on the e-house, was renovating a bathroom for a nearby homeowner. (The consensus best part of that project so far: demolishing the existing fixtures.) "We charge about a fifth of what a contractor would, but the jobs take longer," McCartney said. "Homeowners are sometimes hesitant, at first, when they see a school bus pull up with eighteen kids. But by the end they know all the kids' names, and they've often fed them on numerous days." Juniors and seniors can also leave school for paid part-time jobs, many of which become full-time after graduation.



The electrical students Gabe Green (lower left) and Thomas Yulo (lower right) practice wiring a wood-framed wall in their school's shop.

McCartney and I walked down the hill to a paved open space, where electrical students were cutting lengths of metal conduit with hacksaws, then using hand tools called benders to create six-inch offsets. Inside the building, Aleena Rivera, a sophomore, presented her conduit to the instructor, who held it up to a steel junction box on the counter in front of him, next to a tape measure and a hardcover copy of the National Electrical Code. “Look at that,” he said. “That is beautiful.” She smiled. The shop is so large that students are able to repeatedly build and rebuild a two-story structure entirely inside it, with residential wood framing on the ground floor and commercial steel framing above. (Electricians need to know some carpentry, too.) They practice wiring both floors, each in accordance with the code for its construction type.

I followed two juniors, Gabe Green and Thomas Yulo, to the second floor. “We were downstairs last year,” Yulo said. “So we replaced most of the wood studs down there.” Green was wearing a gray hoodie and Yulo a long-sleeved gray T-shirt, and each had the word “electrical” embroidered in yellow over the left breast and the boy’s name over the right. “When I came here, I was thinking about machining,” Yulo said. “But then I thought, Do I really want to be working inside all my life?” Green said, “I’ve got uncles and aunts in trades. My dad is a carpenter. He does ductwork and all of that, and my brother is a plumber, and my other brother does *HVAC*. My father wants us to be, like, a whole family business.” They showed me their theory classroom, a large, bright space that opens off of the back of the shop. There were eighteen desktop computers and big stacks of books, mostly related to the electrical code. Yulo said, “A lot of kids nowadays, they’re not into all the textbooks, but when we get hands-on, now we can understand, and we’re more interested in what we’re learning about.” By the time they graduate, they’ll have logged seven hundred and twenty hours of theory, which is all the class work they need for their licenses.

Students I met at all three schools were aware of and interested in the environmental impact of the skills they were learning, but none of them, as far as I know, had applied to trade school because they were worried about climate change. Like Green and Yulo, they were following the example of family members who worked in similar trades, or they or their parents were alarmed by the cost of college, or the *COVID* shutdown had soured them permanently on spending all day staring at a screen. But to create the kind of green workforce that Bill McKibben envisions you don’t need to turn anyone into a McKibbenite. One reason Republicans are unlikely to gut the I.R.A. is that its focus is on infrastructure and job creation, rather than on changing anyone’s mind about the causes of global warming.

Connecticut’s public-trade-school system is similarly resistant to ideological meddling, because the curriculum is tied directly to the regional job market. The I.R.A. has altered the incentives for many kinds of public and private investment, and the schools are nimble enough to respond quickly. Dr. Ellen Solek, the system’s executive director, told me, “Electric vehicles are coming out like crazy, but we don’t have enough charging stations, and we don’t have a preponderance of people trained in installing high-speed chargers. So guess where we’re going in our automotive programs?” Solek spent sixteen years as a music teacher and twelve as a middle-school principal, and *CTECS* is the third school district she’s headed as superintendent. “I remember when all the auto-body shops and home-economics shops in academic public schools

were being shut down, back when I was teaching," she said. "Why? Because technology was coming, and we needed computer labs instead. Now there's been a one-eighty."

My internist, my dermatologist, my gastroenterologist, my dentist, my veterinarian, and all four of the Navy pilots who flew fighter jets over this year's Super Bowl are women. My lawyer is a man, but last spring—and I'm not saying that this proves anything about men generally—he was charged with first-degree manslaughter, after shooting someone in the parking lot of one of his firm's offices. (He pleaded not guilty.) More women than men now go to college, and they have better grades and a higher graduation rate. Since 2014, they've also been more likely to go to law school, and since 2019 they've been more likely to go to medical school.

Men still dominate the heavy-duty skilled trades. The gender disparity partly reflects the physical nature of the work, but there's a cultural element, too. Last year, my wife and I decided that we needed central air-conditioning, after surviving thirty-six New England summers without even a window unit, and we had a heat-pump system installed at our house. (That system has turned out to be surprisingly effective at heating the house, too, even when the outside temperature is far below freezing. Someday we'll get rid of our oil-fired boiler, which we now use mainly as a backstop.) The crew that did the work was all men. Gary Pelletier, who owns the company, told me that female techs are a rarity in his field, but that he's hired a couple in the past. "I had a customer once who complained that a woman had no business doing this," he said. "But a lot of customers welcome it, and it's an up-and-coming thing."

Total enrollment in Connecticut's public-trade-school system is about eleven thousand. Girls account for forty per cent of the total. They disproportionately choose fields such as culinary arts, hairdressing and cosmetology, graphic technology, guest-services management, and health, but their interest in traditionally male-dominated trades is growing, partly because demand from employers is high and rising. They now make up roughly a quarter of the carpentry students, a fifth of the electrical students, a third of the masonry students, and a sixth of the plumbing and heating students.

When I visited Howell Cheney Tech, in Manchester, another school in the system, Jousette Caraballo, the dean of students, pointed to two framed photographs on a wall in the main office. In each of them, a young woman was standing on a stepladder and working on the exposed interior of a jet engine, which appeared to be the size of a small house. "That's me, a little over twenty years ago," she said. "I grew up in the Bronx. I'd look up and see all these giant jumbo jets flying overhead and wonder, How are they staying up there?" In a conference room down the hall, a (male) culinary-arts instructor laid out a feast that students had prepared that morning: scones, pastries, coffee cake, double-chocolate brownies, muffins, quiche. Caraballo and I were joined by Hadley Gonzalez, a senior carpentry student. Last year, Gonzalez completed an eight-week paid internship at General Dynamics Electric Boat, the country's leading manufacturer of submarines, based in Groton. Two weeks before the internship ended, the company offered her a full-time job. I asked her what a carpenter does on a submarine. "We do exterior work, interior work, soundproofing," she said. "We build the scaffolding you need to get on top of the boats, and we make sure that everything is round and flat, so that it's safe." She apologized for sounding vague; some of the projects she worked on could be classified.

Gonzalez has long brown hair, which hangs past her waist, and she pulled it out of the way so that I could see the logo on the back of her hoodie: a Columbia-class nuclear submarine, an American flag, and a trident, encircled by a gold braid and the motto “The Future of Strategic Deterrence.” (It was a parting gift from her co-workers.) She told me that her interest in carpentry had come from her grandfather. “He was a truck driver, but we built a shed together and did a lot of little projects around the house,” she said. “I liked creating stuff with my hands.”

After breakfast, Gonzalez took me to Cheney’s carpentry facility. There were industrial-size saws, jointers, routers, planers, shapers, sanders, and other tools, plus many orderly stacks of lumber and an extensive built-in dust-collection system (which is maintained by electrical students). “This is my home, and I love it,” she said. “I don’t care what culinary says; carpentry has the best-smelling shop.” Her teacher was on his way out the door. He was taking fifteen juniors to work on a construction project at a nearby house. “We’re doing a deck with railings and a hip roof,” he said. “It’s a nice little job.” Gonzalez showed me something that she and her classmates had been working on inside the shop: an old, wooden, horse-drawn carriage, which they’re restoring for its owner. Several critical metal parts had been missing, so students in precision machining had fabricated replacements.



Precision machining at Kaynor.



Welding at Howell Cheney.

Stanley Black & Decker, the country's largest tool manufacturer, is based in New Britain, less than twenty miles from the Cheney campus. The president of the company's power-tool group is

Allison Nicolaidis, who, like Hadley Gonzalez, was introduced to amateur tinkering by her grandfather. I asked her whether the country had enough skilled workers to fully implement the I.R.A. She said, “If you asked any of our folks who are running the kinds of large companies that tend to win those contracts, they would say no.” Last year, Stanley Black & Decker published a report, called the Makers Index, which estimated that there were six hundred and fifty thousand unfilled jobs in construction-related trades in the United States, and ten million worldwide.

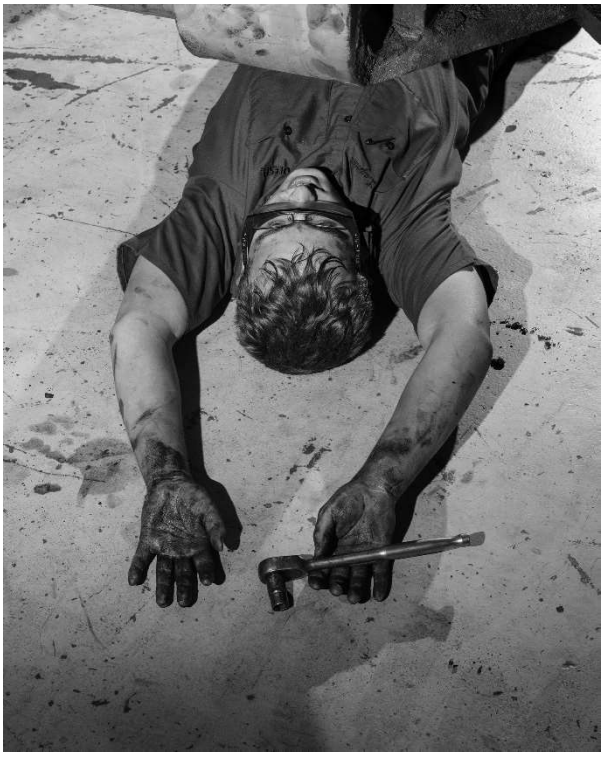
Many formerly daunting jobs have been made more accessible by changes in technology. Some types of commercial construction now employ a form of prefabrication, called “manufacturization,” in which tasks that used to be done exclusively on-site are performed inside huge, climate-controlled spaces that are equipped like factories. “When you do that, you can use equipment that you could never have on a job site,” she said. “You can build a twenty-foot run of mechanical, electrical, and plumbing, all on one big rack, then send it to the site on the back of a flatbed, with a tag that tells the installing crew where to plug it in.” Tools are evolving, too. “Think of something like an impact wrench, which is a high-powered fastening tool that you use to drive big bolts,” she said. “Twenty-five years ago, when I started, it was absolutely a corded tool, and it was as heavy as a bowling ball. Now it’s cordless, and it weighs a third as much.” These and other changes have been good for both men and women: lighter tools and less exposure to the elements make for fewer injuries and longer careers.



Nearly all of Connecticut’s trade apprentices come from *CTECS*, which has about eleven thousand students. The curriculum is a “fluid document” that evolves alongside policy changes and labor demand.

Gary Pelletier, whose company installed the heat pump at my house, said that representatives from Mitsubishi, which manufactured my unit, have assured him that demand will be strong, worldwide, for at least twenty-five years. “They told us that we should just be worrying about where our people are going to come from,” he said. (Last year, in the United States, heat pumps overtook gas furnaces in total sales; in Europe, installations were up almost forty per cent over 2021, partly as a result of efforts to reduce the importation of Russian natural gas.) Pelletier took me to see a big air-conditioning job that his crew had been working on, in a thirty-year-old house whose owners, like my wife and me, had decided that New England summers were now too hot. An apprentice, working in the garage, was sealing the joints on a new length of ductwork. The electrical subcontractor who had worked on my job was connecting something to the main service panel, in the basement. The owner of the house was typing at a computer on her dining-room table and trying to ignore the commotion all around her.

Andrew Cozza, who manages Pelletier’s installation department, was working in a room on the second floor which appeared to be either an on-site self-storage unit, now approaching capacity, or a home gym. Cozza is thirty-eight years old and has an impressive portfolio of tattoos. Like his boss, he went to Oliver Wolcott Tech, in Torrington, another *CTECS* school, but studied culinary arts. He joined the Marine Corps after graduation, and when his tour ended, four years later, he got a job in a factory. “I bought a house as soon as I got discharged,” he said. “One morning, I woke up to no heat, and when the service technician got it going again I was, like, Wow, that guy is my superhero.” He decided he would like to have a job like that, too, and so signed up for night classes at a private tech school, in Watertown, and fell in love with *HVAC*. “It sounds crazy to say this, but my job doesn’t even seem like work to me,” he said. “The sky’s the limit right now. Once you’ve got your mechanic’s license, you can easily make six figures.”



A student at Howell Cheney works on a tractor during class.

Leah Stokes, the U.C. Santa Barbara professor, told me, “For a long time, we have valorized white-collar jobs and tech workers and the knowledge economy. We need a whole new group of people to think about going into the trades, including people whose families have had white-collar jobs.” One of my golf buddies is a pilot for American Airlines. He and his wife have a daughter who’s about to start graduate school, a daughter who’s about to start college, and a son, Sam, who, in addition to having a decent golf swing, is an apprentice at a local *HVAC* company. Sam is twenty. He knew when he was in eighth grade that he didn’t want to go to college. He attended his town’s regular public high school, and, after graduation, went to work. He takes night classes at Henry Abbott Tech, another school in the *CTECS* system, accumulating theory credits toward his journeyperson’s license. He told me that his sister’s bachelor’s degree had cost his parents about two hundred thousand dollars, while the twenty credits he needs for his license will end up costing more like five thousand. (*CTECS* night-school students pay tuition.) Meanwhile, he’s installing heat pumps and getting paid.

My wife’s and my first house was built in the late seventeen-hundreds. It was a classic fixer-upper, and I spent thirty-five years fixing it up. As I visited Connecticut’s tech schools, I brooded that classes in carpentry, roofing, cabinetmaking, plumbing, electrical, and *HVAC* would have been far more useful to me, in my life as I have lived it, than all those years of French. I felt a little better later, back in my office, when I realized that the most valuable parts of my actual education had really been vocational, too: writing for and editing school publications, beginning in grade school and continuing all the way through college. I paid much more attention to those activities than I did to any of my classes, and the practical experience I gained led directly to jobs in writing, book publishing, and journalism, both before and after graduation. So I guess my years of college prep (and college) weren’t completely wasted.



CTECS student crews are able to take jobs in the surrounding community. The consensus favorite aspect of a recent bathroom renovation: demolishing the existing fixtures.

Not that long ago, big-name tech companies made headlines by offering their employees extraordinary perks (along with lawyer-size starting salaries): free meals prepared by in-house chefs, months-long paid parental leave, on-site massages and dental care, dry cleaning, haircuts, valet parking, lounges filled with arcade games, all-day snack buffets. More recently, the main tech news has been about job cuts: twenty-one thousand at Facebook's parent company, Meta, twelve thousand at Google, twenty-seven thousand at Amazon—as well as more than half of the entire workforce at Twitter, including a man with a disability who had to get into a public tweet fight with Elon Musk in order to confirm that he'd been fired. And let's not even think about what's happened to English majors.

Meanwhile, Electric Boat has announced that it expects to hire fifty-seven hundred new employees this year, and twenty thousand during the next decade; it's even begun advertising job openings on TV. I recently attended a job fair at Cheney, organized by Jousette Caraballo. There were representatives from construction companies, aerospace manufacturers, trade unions, colleges, electrical contractors, the state apprenticeship council ("The other 4-year degree"), the military, Olive Garden. The director of a regional apprenticeship program of the Laborers'

International Union of North America told me that members of his union, at the end of their apprenticeships, earn more than fifty dollars an hour in wages and benefits. One of his challenges, he said, is overcoming the hesitation of parents who worry that "laborer" means "sweeper."

A few days later, I had dinner with Marc LeMieux, my superhero at Christmas, and his wife, Jamie, at a crowded restaurant in Watertown, a few miles from the campus of W. F. Kaynor, another *CTECS* high school I visited. (We actually could have had lunch at Kaynor, in its hugely popular student-run dining room, if I'd thought to make a reservation far enough in advance.) "I always liked taking stuff apart and putting it back together," Marc said. "The summer after eighth grade, my family's water heater went out. I went downstairs with the plumber and watched him remove the old one and put in the new one, and I was like, I want to do *that*." He graduated from Wolcott—as Pelletier did fourteen years later and Cozza six years after that—and got a job with a local plumber.

Jamie spent two years at a local university and two years at the Art Institute of Pittsburgh, a trade college. She worked as an airbrush and layout artist at a department-store chain, then for thirty years in the printing industry. She still works as an artist, freelance, and also as a classroom assistant at Wolcott.

"And you married a plumber," Marc said.

She and Marc have a daughter, who's about to finish graduate school, in speech pathology, and a son, who's in his second year at the New York School of Interior Design.

"We're two working parents, and it's still hard to afford," Jamie said. "Some of the kids I'm with every day at Wolcott—even if they did want to go to college, there would be no way." ♦